

Netflix Content Analysis: Data Cleaning and Exploratory Insights

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Objective

The main objective of this project is to analyze Netflix's content dataset to understand trends in movies and TV shows. The focus is on data cleaning, exploratory data analysis (EDA), and visualization.

Key goals:

- Clean and prepare raw Netflix dataset
- Perform exploratory data analysis
- Identify content trends
- Analyze ratings, genres, and release patterns
- Build an interactive dashboard for insights

Tools & Technologies Used

- Python (Pandas, NumPy, Matplotlib) - Data cleaning & EDA
- SQL - Data querying & aggregation
- Excel - Initial data inspection & cleaning
- Power BI - Interactive dashboard & visualization

Dataset Overview & Preparation:

The project used two separate datasets (Movies and TV Shows), which were individually cleaned by handling missing values and standardizing formats. After preprocessing, both datasets were merged into a single dataset for unified analysis.

The database contains the following information

- **show_id**: Unique identifier for each title
- **type**: Movie or TV Show
- **title**: Name of the content
- **director**: Director of the show/movie
- **cast**: Main actors
- **country**: Country of production

- **duration:** Runtime (minutes for movies / number of seasons for TV shows)
- **genre:** Category of content
- **description:** Brief summary of the content
- **popularity:** Popularity score of the title
- **vote_count:** Number of user votes
- **vote_average:** Average rating
- **budget:** Production budget
- **revenue:** Revenue generated
- **budget_m:** Budget in millions
- **revenue_m:** Revenue in millions

Data Cleaning Process

1. Initial Preparation

The project started with two datasets (Movies and TV Shows). Columns were aligned and standardised across both datasets to ensure consistency in structure.

2. Handling Missing Values

Missing values were identified in columns such as director, cast, and country, and were replaced with appropriate null values for consistent analysis

3. Data Formatting

Data types and text formats were standardized across both datasets. This included ensuring uniform formatting for categorical fields like type, genre, and country.

4. Final Preparation

After cleaning and formatting, both datasets were made structurally consistent and ready for merging into a single dataset for analysis.

Python Analysis (Jupyter Notebook)

Data Import & Merging

The cleaned datasets were imported into Python (Jupyter Notebook) from Excel. Both the movies and TV Shows datasets were merged into a single dataframe and verified .

Exploratory Data Analysis (EDA)

After merging, exploratory analysis was performed to understand content trends. Key insights were derived by analyzing:

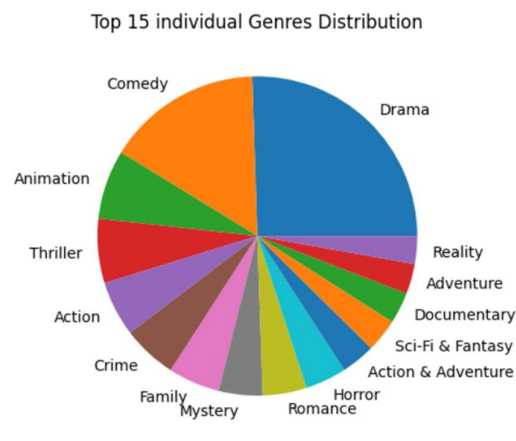
1. Country-wise Content Distribution

- The dataset was analyzed to identify countries producing the most Netflix content.
- Since country values were stored in mixed formats, data was standardized before aggregation.
- **Top country:** USA with 7,760 releases followed by Japan with 2,587 releases

Insight: Netflix content production is heavily concentrated in the USA compared to other countries.

2. Genre Analysis

- Genres were stored in mixed formats and were cleaned for proper separation.
- Frequency analysis was performed to identify dominant genres.
- **Top genres:** Drama followed by Comedy



Insight: Drama and Comedy dominate Netflix content, indicating strong audience preference for these categories.

3. Rating Analysis

- Separate analysis was performed for:
 - Movies
 - TV Shows
 - Overall content
- It was observed that overall top-rated content does not always match top-rated movies or TV shows individually as multiple shows can have top rating

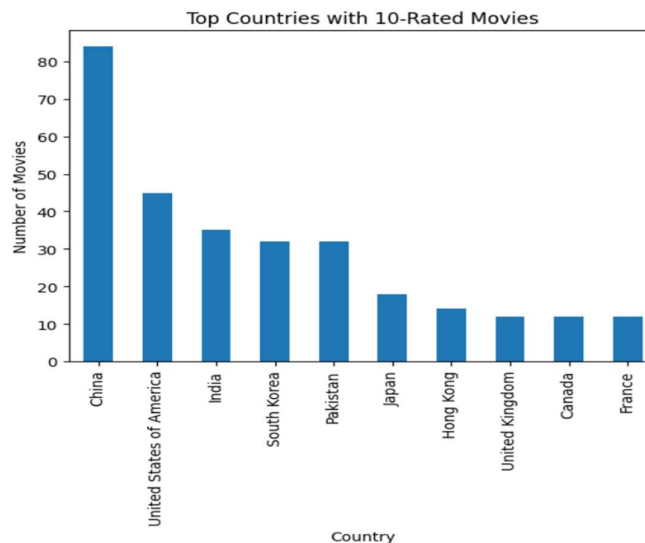
4. Profitability Analysis

- Revenue and budget data were analyzed to identify highly profitable titles.
- Profit was calculated using revenue-based comparison.
- **Top profitable title:** *Avengers: Endgame* - 2.44e+09 profit

5. High Rating Content by Country

- Analysis was performed on content with perfect ratings (10/10).
- **Top countries producing 10-rated shows:** China with 81 shows followed by USA with 35 shows

Insight: While the USA dominates in volume, some countries show higher concentration of perfect-rated content.



6. Genre vs Rating Analysis

- Average rating was calculated for each genre.
- **Top genres by mean rating**
 - Action & Adventure - 6.76
 - Sci-Fi & Fantasy - 6.74
 - Animation - 6.70

Insight: Certain high-production genres also maintain relatively higher audience ratings.

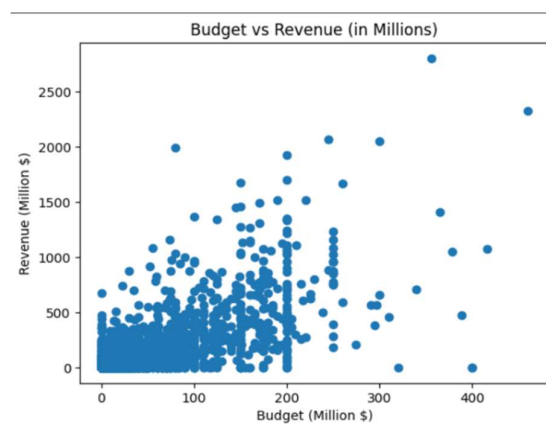
7. Content Description Analysis

- Description lengths were analyzed to identify patterns in content summaries.
- Titles with shortest descriptions:
 - *Alahed* - 6 characters
 - *The Big Slime Movie* - 7 characters
 - *Lady Night* – 8 characters

8. Revenue vs Budget Analysis (Visualization)

- A scatter plot was created to analyze relationship between budget and revenue.
- Most content is concentrated in: Low budget – low revenue region

Insight: High budget does not guarantee high revenue; most titles fall into low investment, low return category.



Data Export to SQL

After analysis, the final merged dataset was exported from Python to SQL for further querying and structured analysis.

SQL Analysis

After cleaning and preprocessing the dataset in Python, the final merged dataset was loaded into SQL for structured querying and deeper analysis.

1. Movies vs TV Shows Trend

- The number of Movies and TV Shows was compared across different years.
- The dataset shows a nearly equal distribution of Movies and TV Shows each year.
- No clear dominance of either category is observed over time.

Insight : The dataset reflects a balanced content distribution between Movies and TV Shows.

2. Genre Distribution Analysis

- Genres were initially stored in combined/mixed formats and were split for accurate analysis.
- Top genres identified are Drama with 14,776 occurrences and Comedy with 9,110 occurrences

Insight: Drama is the most dominant genre, followed by Comedy

3. Budget vs Rating Analysis

Movies were categorized based on budget and rating to study their relationship

High budget – High rating - 1252	High budget – Low rating - 3569
Low budget – High rating - 2649	Low budget – Low rating - 8530

Insight: Budget does not strongly determine rating outcomes. A large portion of content falls into the low-rating categories regardless of budget level.

4. Country-wise Rating Analysis

Top countries producing highly rated content were identified based on average ratings , The three countries being

Pakistan - 7.61

Qatar - 7.08

Cuba - 6.70

Insight: Smaller content-producing countries show relatively high average ratings, suggesting quality focused output despite lower volume.

5. Profitability Analysis

- Profit-to-budget ratio was calculated and identified top 3 most profitable titles and bottom 3 least profitable titles

Insights : “The Beatles: Eight Days a Week - The Touring Years” has the highest profit-to-budget ratio and “The Good Doctor” with the least ratio

6. Director Analysis

Director with the highest number of Netflix shows and found Tyler Perry has directed 20 shows in Netflix

7. Actor Participation Analysis

Analysis performed to check actor involvement across Movies and TV Shows.

Jackie Chan appeared in 23 Movies and 0 TV Shows in Netflix

Insight: Actor involvement is strongly skewed toward movies for Jackie Chan

8. Genre vs Budget Analysis

- Genres were analyzed based on average budget allocation.

Top 3 budget-heavy genres were

- Adventure
- Science Fiction
- Fantasy

Insight: High-budget production is concentrated in visually intensive genres such as Adventure , Science Fiction and Fantasy

9. Multi-Genre Content Analysis

- Titles were analyzed based on number of genres assigned.
- Highest multi-genre show being “ *The Voice Kids* ” with 16 genres
- **Insight: Some shows are highly multi-genre, indicating broad categorization and wider audience targeting.**

Power BI Dashboard

A Power BI dashboard was developed using the cleaned dataset to visualize key business and content insights from Netflix data.

1. Key Visualizations

- KPI cards displaying total content count and overall average rating
- Pie chart showing distribution of Movies vs TV Shows
- Clustered bar chart representing dominant languages in Netflix content
- Line chart showing revenue trends over the years
- Pie chart showing top genres contribution (aggregated view)

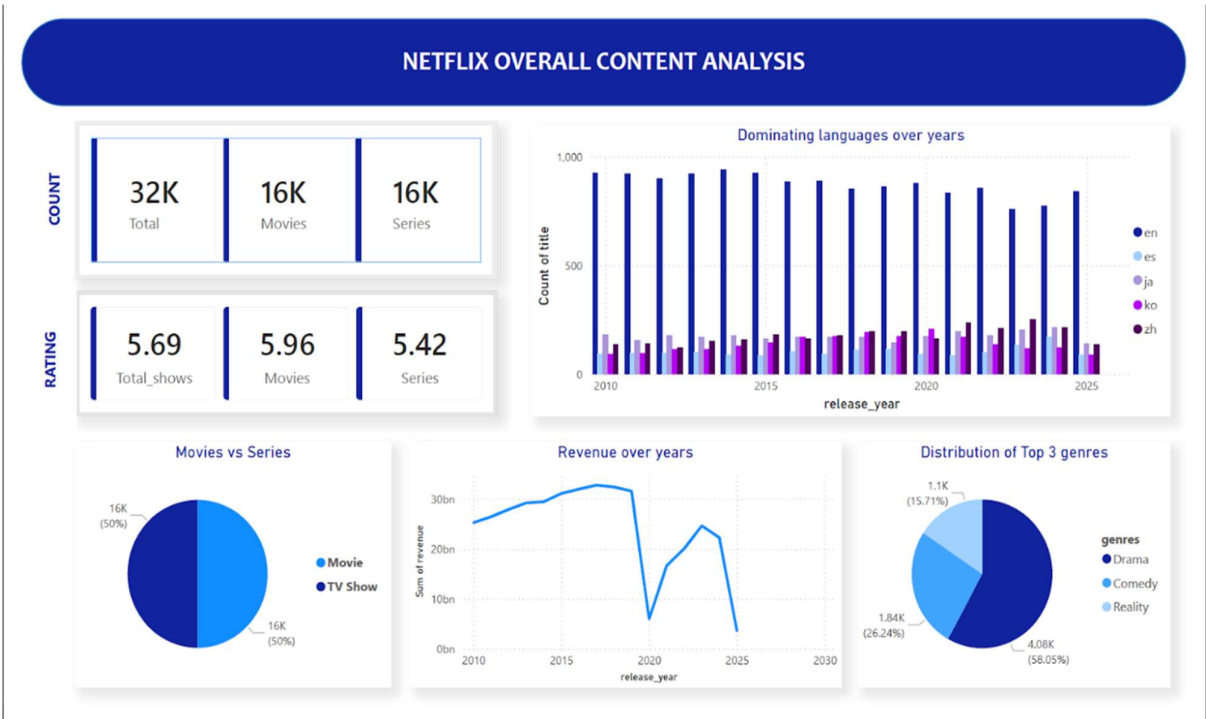
2. Filters and Interactivity

- Content type filter (Movie / TV Show)
- Year-wise filtering for trend-based analysis
- Language-based filtering for regional insights
- Genre filtering for content exploration

3. Dashboard Outcome

The Power BI dashboard provides a clear and interactive view of Netflix content distribution, language dominance, and revenue trends over time. It helps in understanding overall content structure and performance patterns.

Dashboard



Conclusion

This project provided a complete end-to-end analysis of Netflix Movies and TV Shows, starting from raw data cleaning to structured insights generation. Through exploratory analysis, SQL queries, and Power BI dashboards, meaningful patterns were uncovered in content distribution, ratings, language trends, and revenue performance. The project demonstrates practical experience in handling real-world datasets and transforming them into actionable insights through data analysis and visualization.